

Felipe Bedoya

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GitHub: <https://github.com/FbStatsQuant>

Professional Summary

Robust foundation in statistics, applied mathematics, and economics. Extensive experience in Bayesian modeling. PhD candidate in Statistics at Rice University with a focus on advanced Bayesian methods, including nonparametric Bayesian estimation for regression, as well as proximal MCMC for quantile sampling.

Education

PhD in Statistics

Rice University, Houston, TX

(Expected May 2027)

GPA: 3.9

- Research: Nonparametric Bayesian estimation, proximal MCMC, quantile regression and variable selection.
- Relevant Coursework: Stochastic Processes and Applications to Finance, Statistical Machine Learning, Deep Learning, Econometrics, Time Series, Probability Theory.

Master of Arts in Statistics. Rice University, Houston, TX. December, 2026

Master of Science in Applied Mathematics. University Sergio Arboleda, Bogotá, Colombia
Research: Application of bivariate copulas for equity index modeling (2015–2020).

Bachelor of Science in Mathematics.

University Sergio Arboleda, Bogotá, Colombia

Research: The Black-Litterman Model, a non-normal approach.

Bachelor of Science in Economics.

Colombian School of Engineering, Bogotá, Colombia

Research: Biofuels and food inflation in Colombia (2006–2012);

Additional Research: Returns to Education in Colombia (2009–2015).

Experience

Research Assistant

Rice University, Houston, TX

(August 2024 – Present)

- Research in nonparametric Bayesian estimation and proximal MCMC for quantile regression under Dr. Meng Li.
- Uncertainty quantification of Bayesian models.

Graduate Instructor – Probability and Statistics
(August 2024 – December 2024)

Rice University, Houston, TX

Teaching Assistant – Quantitative Financial Risk Management Rice University, Houston, TX
(August 2022 – August 2024)

- Supported instruction in risk analysis, derivative pricing, and portfolio management. Assisted students with Black-Scholes, implied volatility, Greeks analysis, and trading strategies.

Instructor – Microeconomics and Mathematical Economics University Sergio Arboleda, Bogotá, Colombia
(2015 – 2022)

- Taught microeconomics, game theory, and mathematical economics. Applied differential equations and advanced mathematics to economic modeling.

Relevant Projects

- Deep Learning for Volatility Forecasting
- Artificial Neural Network Examples
- Gaussian Process Regression in Python and R
- Bayesian Quantile Trend Filtering via Proximal MCMC
- Bayesian Sparse Function Recovery with Random Series Priors

Research in Progress

- **Large Bayesian Prior: An Adaptive and Optimal Method.** Develops an efficient fully Bayesian method for nonparametric regression that uses large deterministic series priors and global-local shrinkage, adapts to unknown smoothness, and achieves minimax posterior contraction. In preparation for *Biometrika*.
- **Bayesian Joint Quantile Trend Filtering via Proximal MCMC with Sequential Gibbs Sampling.** Develops joint non-crossing quantile trend filtering with adaptively learned smoothness, sequential Gibbs/P-MALA updates, and ADMM projections; simulations show accurate estimates and calibrated uncertainty.
- **Local-to-Global Bayesian Computation.** Develops a divide–merge–correct framework that combines local posterior computations and corrects global candidates through importance weighting, sequential Monte Carlo, tempering, or Metropolis–Hastings, with applications to variable selection and phylogenetics.